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Long-term changes in land surface temperature due to land use land cover over a mega city in south India

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ABSTRACT

It is well known that one of the major causes of global climate change is urbanization and industrialization. The necessity of converting vegetation area into residential and commercial areas and their allied developments in the way of rapid urbanization, may change the surface albedo that could increase the Land Surface Temperature (LST). The main objective of the present study is to evaluate the impact of landscape structure on the diversity of LST over one of the fastest growing urban region Hyderabad, located in Rangareddy district of Telangana state in southern India. Study has been carried out during the period of 1995 - 2018, using four Landsat Images (two Landsat - 5 Thematic Mapper and two Landsat OLI-TIRS 8). Land Use Land Cover (LULC) mapping and LST are derived by using pre-processed landscape data. Interestingly, urban region LST is found less than that observed in the outskirts during entire study period. Land cover classification showed that the built-up areas, barren land and vegetation increased by 2.34%, 9.28% and 18.01%, respectively, while the forest area and water bodies are decreased by 28.79% and 1.21% respectively, over the period of 24 years between 1995 and 2018. Maximum LST for the total area increased by approximately 4°C and the minimum temperature increased by approximately 3.5°C between 1995 and 2018. The negative correlation has been observed between Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) with LST with low temperatures, but Normalized Difference Built-up Index (NDBI), has positive correlation with high temperatures. This study also provides a quantitative approach in exploring the relationship between several index and temperature.

Keywords: Land Use Land Cover (LULC), Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), Normalized Difference Water Index (NDWI), Arc GIS.

INTRODUCTION

Rapid urbanization has led to continuous decline in greenery in most of the cities. Due to population growth and urban sprawl, the vegetation landscape is undergoing unwanted changes. The major reduction in the forest and agricultural lands, and an increase in barren land and impermeable surfaces due to huge built-up areas, are the main reasons for the observed changes in Land Use Land Cover (LULC) (Kumar et al., 2012). Landscape architecture incorporates human and natural components and their spatial pattern (Dissanayake et al., 2019). In the day time, buildings and non-vegetative surface, such as bare soil, can trap the incoming solar radiation (Staniec and Nowak, 2016; Weng et al., 2004), and then re-radiate during the night time, due to the absence of solar insolation (Arsiso et al., 2018). At present, urban areas suffer from the most critical problem in loss of vegetation areas, thereby increase in surface temperatures due to uncontrolled transport, evaporation, and the growth of hard ground surfaces (Buyadi et al., 2013; Hussain et al., 2014; Kant et al., 2009; Kumar et al., 2012). The regional and global impact of changes in terrestrial ecosystems, due to human activities through change in LULC, can be clearly monitored using the thermal infrared (TIR) remote sensing energy, which is emitted directly from the land surface (Faqe Ibrahim, 2017).

The Normalized Difference Vegetation Index (NDVI) is the most commonly used vegetation index for monitoring vegetation and identify different vegetation cover worldwide. However, in an area with low plant cover (urban areas), soil and climate have a significant impact on NDVI and its relationships with vegetation (Huete and Jackson, 1987). It is imperative to quickly and accurately describe the water-coverage area in order to dynamically monitor water resources and planning water conservation. The most commonly used method for water characteristic extraction, is the Normalized Difference Water Index (NDWI) proposed by Mcfeeters (1996). The NDWI produces negative values for vegetation and soil and positive values for water bodies. The Normalized Difference Built-up Index (NDBI) is an effective indexing mechanism used to automatically mapping urban built-up areas on Thematic Mapper (TM) images. The built-up area appends all impervious surfaces, including houses, industries, roads, etc. This method takes upper hand of the spatial spectral responses of built-up areas and other land covers. Remote Sensing (RS) and Geographical Information System (GIS) techniques are used to detect the land-use changes and its impact on the land surface temperature (Buyadi et al., 2013).

In an earlier study, Rao (1972) introduced a thermal infrared remote sensing-based LST-estimation method. Later, Lombardo, (1985) developed this concept by developing a computational model for remote sensing-based LST measurements. Since then, various researchers have performed several qualitative studies using different satellite data sets (AVHRR/NOVAA/ASTER/Landsat(TM/ETM+)/MODIS TIR data set) and some of them reviewed and suggested a satellite view of the LST's assessment and its relationship to